

Jike Wang (王纪科)

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EDUCATION

Shanghai Jiao Tong University, Shanghai, China

Sep. 2021 - Jun. 2026

Ph.D. in Computer Science and Technology

Research Interests: Artificial Internet of Things (AIoT), Wireless Sensing, and Mobile Robotics

Advisor: [Prof. Dongyao Chen](#)

Xidian University, Shaanxi, China

Sep. 2017 - Jun. 2021

B.S. in Software Engineering

GPA: 3.9/4.0

Rank: 3/459 (Top 1%)

RESEARCH OVERVIEW

My research is driven by the goal of enabling reliable perception and autonomy for intelligent systems in the real world. I focus on building robust, low-power sensing systems for intelligent robots and AIoT platforms that can operate effectively in complex environments where conventional vision-based approaches often break down. By combining wireless sensing, mobile robotics, and embedded intelligence, I aim to expand the sensing frontier for real-world autonomy under occlusion, interference, and resource constraints.

A central theme of my work is *magnetic sensing*, a modality inherently resilient to occlusion, lighting variation, and wireless interference. Along this direction, our team developed **METRO** for all-weather road-marking perception, **Polaris**, a vision-free magnetic fiducial system for accurate robot pose estimation, and **MagLens** for mobile fine-grained imaging of hidden ferrous structures. Our work has appeared at leading international venues, including ACM MobiCom, ACM SenSys, and NDSS Symposium.

PUBLICATIONS

- **MagLens: Bringing Mobile, Fine-Grained Imaging to Ferrous Building Structures**
[Jike Wang](#), Yasha Iravantchi, Mingke Wang, Alanson Sample, Kang G. Shin, Xinbing Wang, and Dongyao Chen
Proceedings of the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (**SenSys 2026**)
- **Polaris: Accurate, Vision-free Fiducials for Mobile Robots with Magnetic Constellation**
[Jike Wang](#), Yasha Iravantchi, Alanson Sample, Kang G. Shin, Xinbing Wang, and Dongyao Chen
Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (**MobiCom 2024**)
- **METRO: Magnetic Road Markings for All-weather, Smart Roads**
[Jike Wang](#), Shanmu Wang, Yasha Iravantchi, Mingke Wang, Alanson Sample, Kang G. Shin, Xinbing Wang, Chenghu Zhou, and Dongyao Chen
Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems (**SenSys 2023**)

- **DualStrike: Accurate, Real-time Eavesdropping and Injection of Keystrokes on Commodity Keyboards**
Xiaomeng Chen, Jike Wang, Zhenyu Chen, Alfred Chen, Xinbing Wang, and Dongyao Chen
The Network and Distributed System Security Symposium 2026 (**NDSS 2026**)
- **OralSense: Versatile, Accurate Oral Sensing With Customizable Magnets**
Zhenyu Chen, Jike Wang, Ye Huang, Xiaomeng Chen, Xinbing Wang, and Dongyao Chen
IEEE Transactions on Mobile Computing (TMC 2026)
- **μ Touch: Enabling Accurate, Lightweight Self-Touch Sensing with Passive Magnets**
Siyuan Wang, Ke Li, Jingyuan Huang, Jike Wang, Cheng Zhang, Alanson Sample, and Dongyao Chen
Proceedings of the 24th IEEE International Conference on Pervasive Computing and Communications (**PerCom 2026**)
- **FreeLoop: Enhancing Traffic Safety and Efficiency by Replacing Left Turns with Right-turning Loops**
Jike Wang, Xinbing Wang, and Dongyao Chen
ACM TURC 2026
- **Demo: Implementation and Benchmark of Magnetic Tracking on Mobile Platforms**
Zhenyu Chen, Jike Wang, and Dongyao Chen
Proceedings of the 2024 Workshop on Adaptive AIoT Systems (**AdaAIoTSys 2024**)

PROJECTS

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| <p>MagLens: Bringing Mobile, Fine-Grained Imaging to Ferrous Building Structures</p> <ul style="list-style-type: none"> • Background: Existing non-destructive inspection tools for hidden ferrous infrastructure (e.g., steel rebars) provide coarse detection results, and are unable to recover detailed geometry in a low-cost and mobile manner. • Contribution: Developed MagLens, a mobile and low-cost magnetic imaging system for hidden ferrous structures, enabling fine-grained contour reconstruction and depth estimation with errors below 1 mm and 0.5 cm, respectively. • Outcome: MagLens was accepted by ACM/IEEE SenSys 2026. | <p>Jan. 2025 – Feb. 2026</p> |
| <p>Polaris: Accurate, Vision-free Fiducials for Mobile Robots with Magnetic Constellation</p> <ul style="list-style-type: none"> • Background: Traditional fiducial systems (e.g., AprilTag) are vulnerable to occlusion, energy consumption, and privacy issues, making them unsuitable for robust robot perception in complex environments. • Contribution: Proposed Polaris, which achieves sub-millimeter pose estimation accuracy ($< 1 \text{ mm} / 1^\circ$) and low power deployment (25.08 mW) based on magnetic sensing. • Outcome: Polaris was published at ACM MobiCom 2024, and received Artifact Available and Functional badges. | <p>Jul. 2023 – Dec 2024</p> |
| <p>METRO: Magnetic Road Markings for All-weather, Smart Roads</p> <ul style="list-style-type: none"> • Background: Visual road surface markings (e.g., arrows) degrade under rain, glare, occlusion, and wear, becoming a major obstacle to reliable and safe driving. • Contribution: Developed METRO, the first magnetic road marking solution with long-term real-world deployment and robust perception (over 96%) under high-speed and challenging conditions. • Outcome: METRO was published at ACM SenSys 2023. | <p>Oct. 2021 – May 2023</p> |

PRESENTATIONS

“MagLens: Bringing Mobile, Fine-Grained Imaging to Ferrous Building Structures”

ACM/IEEE SenSys 2026, Saint-Malo, France

May 2026

“Enabling Robust and Scalable Robot Perception with Magnetic Sensing” (Best Presentation Award)

The 3rd AIoT Sys 2025, Lanzhou, China

Aug. 2025

“Polaris: Accurate, Vision-free Fiducials for Mobile Robots with Magnetic Constellation”

ACM MobiCom 2024, Washington, D.C., USA

Nov. 2024

Microsoft Research Asia (MSRA, Shanghai), China

Nov. 2024

“METRO: Magnetic Road Markings for All-weather, Smart Roads”

ACM SenSys 2023, Istanbul, Türkiye

Nov. 2023

AWARDS

CPS-IoT Week 2026 SIGBED Student Travel Award (\$2,000)

ACM SIGBED | 2026

Second Prize, Outstanding Ph.D. Academic Forum

SJTU, John Center | 2025

Best Presentation Award, Rising Star Forum

The 3rd AIoT Sys Conference | 2025

National Scholarship (Ph.D.)

Ministry of Education of China | 2024

First Prize, Outstanding Ph.D. Academic Forum

SJTU, John Center | 2024

National Encouragement Scholarship

Ministry of Education of China | 2019

National Encouragement Scholarship

Ministry of Education of China | 2018

ACADEMIC/TEACHING SERVICES

Reviewer, IEEE TMC (Transactions on Mobile Computing)

2026

Member, Artifact Evaluation Committee, ACM MobiCom

2025

Reviewer, ACM IUI (Intelligent User Interfaces)

2025, 2026

Teaching assistant: CS1501, Programming Methodology

2021 – 2025 Fall

CS1501 is the introductory course to computer science and programming at SJTU;

Received an **A grade** evaluation, 2022 Fall. Evaluated by 86 students.

Teaching assistant: CS106, Programming Practice

2022 Summer, 2023 Summer